AI-Lingo

Contextual Language Immersion App

Risk Assessment & Mitigation Plan and Key Performance Indicators (KPIs)

Project period: 5 October – 1 December 2026. Initial version; targets are estimates to be reviewed after the first prototype.

1. Project Risk Register

Each risk is rated by probability and impact. Risk level: Critical (act immediately), High (monitor closely), Medium (manage routinely).

ID	Risk	Probability	Impact	Risk Level	Mitigation
R01	AI generates inappropriate/difficult vocabulary	High	High	**Critical**	Check AI output against the user's vocabulary list; regenerate if too many new words; use constrained prompts
R02	AI response takes too long	Medium	High	**High**	Response streaming; timeout with automatic retry; clear loading indicator
R03	AI service/API becomes unavailable	Medium	High	**High**	Clear error message with retry button; saved dialogues for the demo
R04	Internet connection is unavailable	High	Medium	**High**	Store lessons and vocabulary locally in Hive; sync with Firestore when connection returns
R05	AI generates incorrect language content	Medium	High	**High**	Prompt chaining; test with 30–50 prompts; structured JSON output; "Report incorrect answer" button
R06	User learning data is exposed	Low	Very High	**Critical**	Firebase AI Logic (no API keys in the app); strict Firestore Security Rules; no secrets in GitHub
R07	Loss/corruption of user progress	Medium	High	**High**	Cloud backup in Firestore alongside Hive; "last write wins" sync
R08	App crashes during conversation	Medium	High	**High**	Error handling without app termination; unit tests for Blocs; Firebase Crashlytics

ID	Risk	Probability	Impact	Risk Level	Mitigation
R09	Scope becomes too large	High	High	**Critical**	Freeze MVP using MoSCoW; new features need team approval; voice and gamification moved to Future Work
R10	Integration problems between modules	Medium	Medium	**Medium**	Early Firebase/Gemini proof of concept (week 1); feature branches with pull requests
R11	Poor user experience / confusing UI	Medium	Medium	**Medium**	Wireframes before implementation; quick usability test with 5 users
R12	Team member dependency / delayed tasks	Medium	High	**High**	Task tracking in GitHub Issues; code reviews; reassign tasks if a member is unavailable
R13	Implementation phase delay (only 18 days, overlaps with testing and report)	High	High	**Critical**	Start prototype during design phase; postpone Should/Could features; use 30 Nov–1 Dec buffer
R14	Insufficient testing time	High	High	**Critical**	Write tests during implementation; focus on the critical path (Auth → Chat → Vocabulary)

Risk Summary

Risk Level	Count	Risks
Critical	5	R01, R06, R09, R13, R14
High	7	R02, R03, R04, R05, R07, R08, R12
Medium	2	R10, R11
Total	14	

Risk Monitoring

- **Register:** Keep the register in a shared sheet or GitHub Project and update it at every milestone.
- **Review:** Hold a short weekly review of the top 5 risks.
- **Escalation:** Escalate any Critical risk to the project leader immediately.
- **Deferred risks:** Quota limits, privacy compliance, Android fragmentation, and demo failure were deferred and should be revisited in the testing phase.

2. Key Performance Indicators (KPIs)

2.1 Primary KPIs (Focus of the Project)

#	KPI	Measurement	Target	Frequency
1	**AI Response Time**	Average response time	≤ 5 sec	Weekly
2	**AI Request Success Rate**	Successful requests / total requests	≥ 95%	Weekly
3	**AI Content Relevance**	Relevant outputs / tested outputs	≥ 90%	Testing phase
4	**Vocabulary Constraint Compliance**	Compliant outputs / total outputs	≥ 90%	Testing phase
5	**App Crash Rate**	Crashes / sessions	< 2%	Weekly
6	**User Task Completion Rate**	Completed tasks / attempts	≥ 85%	Usability testing
7	**User Satisfaction**	Average survey score	≥ 4/5	After testing

2.2 Supporting KPIs

KPI	Measurement	Target	Frequency
System Uptime	Available time / total time	≥ 99%	During deployment
Error Recovery Rate	Recovered errors / total errors	≥ 80%	Testing
Grammar Correction Accuracy	Correct corrections / reviewed sample (50 responses)	≥ 85%	Testing phase
User Adoption Rate	Active users / test users	≥ 70%	Pilot testing
Lesson Completion Rate	Completed / started lessons	≥ 75%	Pilot testing
Progress Accuracy	Correct progress records / total records	≥ 98%	Testing
Offline Functionality Success	Successful offline tests / total offline tests	≥ 95%	Testing
Critical Security Incidents	Number of incidents	0	Continuous

KPI	Measurement	Target	Frequency
Test Case Pass Rate	Passed / total test cases	≥ 95%	Testing
Must Requirements Completed	Implemented / 12	100%	Each milestone

Notes

- Enable Firebase Crashlytics and Performance Monitoring early so crash and response-time data is collected automatically.
- All targets are initial estimates, especially response time, which depends on the model and network.